LESSON 3.1

Understand Ratios

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write and describe a ratio that compares two quantities.

WHY THIS MATTERS Welcome to Chef Academy! Head Chef Reyes is teaching new students how to read recipes like a pro. Her sparkling cranberry-apple fruit drink calls for 3 cups of apple juice for every 2 cups of sparkling water. Before the students can mix the drink, they need to understand how these ingredient amounts compare to each other.

REMEMBER

Understand Ratios

Understanding Ratios. Formula: a : b | a to b | a/b A ratio compares two amounts. It tells how much of one thing there is for another thing.

1. A ratio compares two quantities; order of quantities matters.
2. Part-to-Part compares two distinct subgroups (e.g. boys to girls).
3. Part-to-Whole compares one subgroup to the total (e.g. boys to class)

Words: **Ratio *(Razón)*** **Comparison *(Comparación)*** **Part-to-part *(Parte a parte)*** **Part-to-whole *(Parte a todo)*** **Colon notation *(Notación con dos puntos)***

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 4 ÷ 1/2?

(A) 1/8
(B) 2
(C) 4 1/2
(D) 8

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 3/4 ÷ 1/2?

(A) 1 1/2
(B) 1/2
(C) 2/3
(D) 3/8

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the reciprocal of 3/4?

(A) 1/4
(B) 3/4
(C) 4/3
(D) 7/4

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

To divide 5/6 ÷ 2/3, what do you multiply 5/6 by?

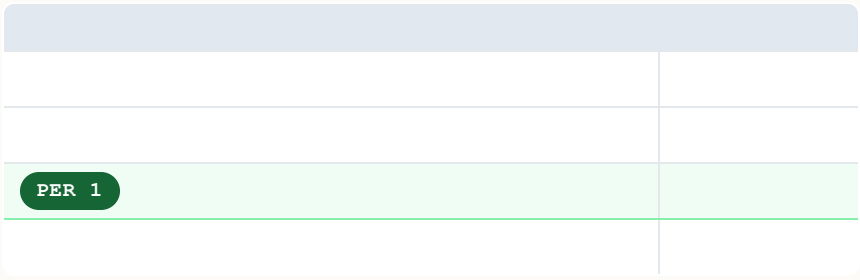
(A) 2/3
(B) 3/2
(C) 5/6
(D) 6/5

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare. Keeping the 3:2 ratio, how many main dishes go with 10 side dishes?

(A) 5
(B) 12
(C) 15
(D) 20



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 **Circle the letter of the best answer. Show how you know.**

The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare. What kind of ratio is 3 mains to 2 sides?

(A) Part-to-whole, because it compares mains to all dishes
(B) A rate, because it uses different units
(C) Part-to-part, because it compares two separate groups
(D) Not a ratio at all

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

SENTENCE STARTER A ratio compares ___ to ___ using words like ___ or symbols like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each kitchen comparison: is it a PART-TO-PART ratio (one part to another part) or a PART-TO-WHOLE ratio (one part to the total)?

Groups: Part-to-Part | Part-to-Whole

_____ 6 cups of broth to 5 cups of noodles

_____ 6 spicy dishes out of 11 total dishes

_____ 8 forks to 3 spoons on the prep table

_____ 3 burnt cookies out of 12 total cookies

_____ 7 savory bites to 5 sweet bites

_____ 4 vegetarian plates out of 10 total plates

2 **Write the letter of the matching item on each line.**

Plating Match: Match each kitchen context to the ratio that describes it in colon form.

_____ 4 lemons for every 3 limes — (A) 4:3

_____ 5 plain rolls out of 9 total rolls — (B) 5:9

_____ 8 clean pans to 1 dirty pan — (C) 8:1

_____ 2 grilled skewers out of 7 total skewers — (D) 2:7

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare. With 15 mains and 10 sides, what is the ratio of mains to TOTAL dishes?

(A) 2 to 5
(B) 3 to 5
(C) 3 to 2
(D) 15 to 10

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Prep Bowl A: Chef Reyes stirs 7 cups of flour with 4 cups of water to make dough. What is the part-to-part ratio of flour to water?

(A) 4:7
(B) 7:4
(C) 7:11
(D) 11:4

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A class has 5 chefs and 7 helpers (12 total). What is the part-to-whole ratio of chefs to the whole class?

(A) 5:12
(B) 5:7
(C) 7:12
(D) 12:5

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

"For every" language: Chef Reyes says, "My glaze uses 9 spoons of honey for every 4 spoons of lemon." Written in colon notation, the ratio of honey to lemon is:

(A) 4:9
(B) 9:4
(C) 9:13
(D) 13:4

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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